

Table 2. Parameter provenance

Panel A. Parameter identity

Symbol	Definition	Value or range	Unit	Status
g_SQ	isotropic doxorubicin semiquinone g value	2.0035	dimensionless	measured
deltaHpp_SQ	peak-to-peak aqueous one-line width	3.25	G	measured
deltaHpp_SQ_replicatio n	independent aqueous one-line width	3.2	G	measured
A_SQ	resolved doxorubicin/adriamycin semiquinone hyperfine values	numeric extraction unavailable	not available	measured source identified
lambda_0	gas-phase O2 spin-spin fine-structure constant	1.9848	cm ⁻¹	fitted
gamma_0	gas-phase O2 spin-rotation constant	-0.0084293	cm ⁻¹	fitted
D_O2_gas	axial D from D=2 lambda convention conversion	3.9695	cm ⁻¹	calculated
D_O2_solid_air	axial matrix O2 zero-field splitting	3.572(3)	cm ⁻¹	measured fitted
D_O2_N2_matrix	O2 zero-field splitting in solid nitrogen	3.497	cm ⁻¹	measured fitted
T1_O2_solution	bulk dissolved oxygen electron spin-lattice relaxation	approximately 7.5	ps	indirectly fitted
K_O2_T4L	O2 association constant in T4 lysozyme L99A cavities	48	M ⁻¹	measured fitted
tau_rot_O2_T4L	O2 rotational correlation components	0.164;1.41	ps	calculated fitted
g_D_E_O2_encounter	O2 g and ZFS tensors in transient doxorubicin-semiquinone/O2 encounter	unavailable	not available	unavailable
T1_T2_Tm_SQ	doxorubicin or close anthracycline semiquinone electron-spin relaxation/dephasing	unavailable	s	unavailable
k_O2_total_pH6	total bulk semiquinone oxidation by O2	3.500e+08	M ⁻¹ s ⁻¹	measured

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Table 2. Parameter provenance

Panel A. Parameter identity (continued)

Symbol	Definition	Value or range	Unit	Status
k_O2_total_pH11_5	total bulk semiquinone oxidation by O2	1.700e+08	M ⁻¹ s ⁻¹	measured
k_O2_total_1981	earlier total bulk semiquinone oxidation by O2	4.400e+07	M ⁻¹ s ⁻¹	measured
t_half_SQ_anaerobic	semiquinone chemical half-life	50	us	measured
diffusion_distance_SQ	author-calculated bulk travel bounds	<0.6 anaerobic;<0.1 air-saturated	um	calculated
encounter_geometry_J_D_kassoc_ksep	separation; orientation; exchange; dipolar; association/separation; cage duration	unavailable	not available	unavailable
DQ_coherence	direct spin-correlated coherent D/Q encounter evidence	unavailable	not available	unavailable
k_D	doublet encounter electron-transfer rate	unavailable	s ⁻¹	unavailable
k_Q	quartet encounter electron-transfer rate	unavailable	s ⁻¹	unavailable
k_Q_over_k_D	quartet/doublet rate ratio	unavailable	dimensionless	unavailable
pKa_HO2	hydroperoxyl acid dissociation constant	4.88	dimensionless	measured
k_HH	elementary HO2 plus HO2 dismutation rate	7.600e+05	M ⁻¹ s ⁻¹	measured
k_HA	elementary HO2 plus O2 radical-anion dismutation rate	8.500e+07	M ⁻¹ s ⁻¹	measured
k_AA	direct superoxide plus superoxide disproportionation upper bound	<100	M ⁻¹ s ⁻¹	measured upper bound
k_SOD_oxidation	bovine CuZnSOD oxidized-enzyme half-reaction	1.200e+09	M ⁻¹ s ⁻¹	measured
k_SOD_reduction	bovine CuZnSOD reduced-enzyme half-reaction	2.200e+09	M ⁻¹ s ⁻¹	measured

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Panel A. Parameter identity (continued)

Symbol	Definition	Value or range	Unit	Status
k_SOD_overall	overall bovine CuZnSOD-superoxide rate	2.370e+09	M ⁻¹ s ⁻¹	measured
v_O2minus_heart_SMP	superoxide formation rate	control 1.6+/-0.2;doxorubicin 69.6+/-2.7	nmol min ⁻¹ mg ⁻¹	measured mean SE
v_H2O2_heart_SMP	hydrogen peroxide formation rate	control undetectable;d oxorubicin 2.2+/-0.3	nmol min ⁻¹ mg ⁻¹	measured mean SE
v_O2minus_Ehrlich_micr osomes	SOD-inhibitable superoxide proxy rate; control vs drug	0.51+/-0.26 n=3;14.71+/-1.43 n=6	nmol min ⁻¹ mg ⁻¹	measured mean SE
v_O2minus_Ehrlich_mito chondria	SOD-inhibitable superoxide proxy rate; control vs drug	1.12+/-0.15 n=7;7.29+/-0.61 n=10	nmol min ⁻¹ mg ⁻¹	measured mean SE
v_O2minus_Ehrlich_nucl ei	SOD-inhibitable superoxide proxy rate; control vs drug	0.36+/-0.05 n=7;3.29+/-0.33 n=12	nmol min ⁻¹ mg ⁻¹	measured mean SE
v_H2O2_Ehrlich	H2O2 formation in microsomes; mitochondria; intact cells	3.42+/-0.48 n=4;2.6+/-0.6 n=3;0.64+/-0.04 n=3	nmol min ⁻¹ mg ⁻¹ ; nmol min ⁻¹ per 1e7 cells	measured mean SE
H2O2_PC3	apparent intracellular steady-state hydrogen peroxide	basal 13+/-4;1 uM doxorubicin 51+/-13	pM	measured derived
MitoSOX_H9c2	relative mitochondrial oxidant fluorescence	1.40+/-0.04;1.80+/- 0.08;2.80+/-0.08	fold control	measured mean SD
negative_5_imino	quinone-disabled anthracycline negative control	doxorubicin 14.71+/- 1.43;5- iminodaunorubicin 0.45+/-0.15	nmol min ⁻¹ mg ⁻¹	measured mean SE

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Table 2. Parameter provenance

Panel B. Chemical conditions

Symbol	Species	Charge state	Environment	Temperature	pH
g_SQ	doxorubicin semiquinone	protonation/formal charge not assigned	N2-purged 100 mM phosphate; 100 uM doxorubicin; 400 uM xanthine; 0.2 U xanthine oxidase in 2 mL	not reported	7.5
deltaHpp_SQ	doxorubicin semiquinone	protonation/formal charge not assigned	N2-purged 100 mM phosphate; 100 uM doxorubicin; 400 uM xanthine; 0.2 U xanthine oxidase in 2 mL	not reported	7.5
deltaHpp_SQ_replication	doxorubicin semiquinone	protonation/formal charge not assigned	hypoxic 50 mM Tris; 1 mM doxorubicin; 400 uM hypoxanthine; 0.1 U/mL xanthine oxidase	310 K	7.4
A_SQ	adriamycin semiquinone	not reported in accessible abstract	not reported in accessible abstract	not reported in accessible abstract	not reported in accessible abstract
lambda_0	16O2 X3Sigma_g- v=0	neutral	gas phase	not extracted	pH not applicable
gamma_0	16O2 X3Sigma_g- v=0	neutral	gas phase	not extracted	pH not applicable
D_O2_gas	16O2 X3Sigma_g- v=0	neutral	gas phase	not extracted	pH not applicable
D_O2_solid_air	ground-state O2	neutral	solid air condensed in waveguide	5 K	pH not applicable
D_O2_N2_matrix	ground-state O2	neutral	O2 diluted in solid N2	15 K	pH not applicable
T1_O2_solution	ground-state O2	neutral	water; 1:1 water/glycerol; cyclohexane; methanol; benzene; acetone; acetonitrile; DMSO	room temperature	not reported/not applicable

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Table 2. Parameter provenance

Panel B. Chemical conditions (continued)

Symbol	Species	Charge state	Environment	Temperature	pH
K_O2_T4L	ground-state O2	neutral	T4 lysozyme L99A; 50 mM phosphate; 25 mM NaCl	298 K	5.5
tau_rot_O2_T4L	ground-state O2	neutral	T4 lysozyme L99A cavities; 50 mM phosphate; 25 mM NaCl	298 K	5.5
g_D_E_O2_encounter	ground-state O2	neutral	doxorubicin-semiquinone solution encounter	unavailable	unavailable
T1_T2_Tm_SQ	doxorubicin semiquinone	unresolved	chemically relevant solution	unavailable	unavailable
k_O2_total_pH6	adriamycin semiquinone plus ground-state O2	biological-pH semiquinone described as internally protonated/zwitterionic; O2 neutral	aqueous phosphate/formate system; O2 or O2/N2 purge at least 30 min	room temperature	6
k_O2_total_pH11_5	adriamycin semiquinone plus ground-state O2	deprotonated semiquinone form; O2 neutral	aqueous borate/OH- system; O2 or O2/N2 purge at least 30 min	room temperature	11.5
k_O2_total_1981	adriamycin semiquinone plus O2	not reported in accessible source	aqueous buffer	not reported in accessible source	not reported in accessible source
t_half_SQ_anaerobic	adriamycin semiquinone	not reported in accessible source	anaerobic aqueous medium	not reported in accessible source	not reported in accessible source
diffusion_distance_SQ	adriamycin semiquinone	not reported in accessible source	aqueous buffer	not reported in accessible source	not reported in accessible source
encounter_geometry_J_D_kassoc_ksep	doxorubicin-semiquinone/O2 encounter	unresolved	chemically relevant solution	unavailable	unavailable
DQ_coherence	doxorubicin-semiquinone/O2 encounter	unresolved	chemically relevant solution	unavailable	unavailable

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Panel B. Chemical conditions (continued)

Symbol	Species	Charge state	Environment	Temperature	pH
k _D	doublet doxorubicin-semiquinone/O ₂ encounter	unresolved	chemically relevant solution	unavailable	unavailable
k _Q	quartet doxorubicin-semiquinone/O ₂ encounter	unresolved	chemically relevant solution	unavailable	unavailable
k _Q /k _D	doxorubicin-semiquinone/O ₂ encounter	unresolved	chemically relevant solution	unavailable	unavailable
pK _a _{HO₂}	HO ₂ radical to O ₂ radical anion	neutral to -1	oxygen-saturated formic acid/sodium formate aqueous buffer	not reported	0-13 investigated
k _{HH}	HO ₂ radical plus HO ₂ radical	neutral plus neutral	oxygen-saturated formic acid/sodium formate aqueous buffer	not reported	0-13 mechanism study
k _{HA}	HO ₂ radical plus superoxide	neutral plus -1	oxygen-saturated formic acid/sodium formate aqueous buffer	not reported	0-13 mechanism study
k _{AA}	superoxide plus superoxide	-1 plus -1	oxygen-saturated formic acid/sodium formate aqueous buffer	not reported	0-13 mechanism study
k _{SOD_oxidation}	bovine CuZnSOD plus superoxide	enzyme state E0 plus -1	aqueous O ₂ ; EDTA; formate	not reported	not reported in accessible record
k _{SOD_reduction}	bovine CuZnSOD plus superoxide	enzyme state E- plus -1	aqueous O ₂ ; EDTA; formate	not reported	not reported in accessible record
k _{SOD_overall}	native bovine CuZnSOD plus superoxide	enzyme plus -1	aqueous	298 K	not reported in accessible abstract

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Table 2. Parameter provenance

Panel B. Chemical conditions (continued)

Symbol	Species	Charge state	Environment	Temperature	pH
v_O2minus_heart_SMP	bovine-heart submitochondrial particles plus doxorubicin	doxorubicin HCl microspecies not reported	isolated submitochondrial particles	not reported in accessible abstract	not reported in accessible abstract
v_H2O2_heart_SMP	bovine-heart submitochondrial particles plus doxorubicin	doxorubicin HCl microspecies not reported	isolated submitochondrial particles	not reported in accessible abstract	not reported in accessible abstract
v_O2minus_Ehrlich_microsomes	Ehrlich-Lette tumor microsomes plus doxorubicin	doxorubicin HCl microspecies not reported	200 ug protein in 1 mL; 150 mM potassium phosphate; 100 uM EDTA; 1 mM NADPH	310 K	7.4
v_O2minus_Ehrlich_mitochondria	Ehrlich tumor mitochondria plus doxorubicin	doxorubicin HCl microspecies not reported	100 ug protein/mL; sucrose/20 mM HEPES/100 uM EDTA; 100 uM NADH; 4 uM rotenone; 5 min preincubation	310 K	8.2
v_O2minus_Ehrlich_nuclei	Ehrlich tumor nuclear fraction plus doxorubicin	doxorubicin HCl microspecies not reported	200 ug nuclear protein; sucrose/20 mM HEPES/100 uM EDTA; 1 mM NADPH	310 K	7.4
v_H2O2_Ehrlich	Ehrlich tumor fractions or intact cells plus doxorubicin	doxorubicin HCl microspecies not reported	fraction buffers as above; intact 1.5e7 cells/3 mL Chelex-treated PBS	310 K	7.4 microsomes; 8.2 mitochondria; cell PBS pH not reported
H2O2_PC3	PC3 human prostate cancer cells plus doxorubicin	doxorubicin microspecies not reported	cell culture	310 K	not reported for intracellular measurement
MitoSOX_H9c2	rat H9c2 cardiomyoblasts plus doxorubicin	doxorubicin microspecies not reported	HBSS with Ca/Mg and 1% BSA; 5 uM MitoSOX	310 K	not reported

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Panel B. Chemical conditions (continued)

Symbol	Species	Charge state	Environment	Temperature	pH
negative_5_imino	Ehrlich microsomes plus anthracycline	dissolved microspecies not reported	microsomal system above	310 K	7.4

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Table 2. Parameter provenance

Panel C. Evidence and permitted use

Symbol	Method	Source	Uncertainty	Limitations	Permitted use	Code location
g_SQ	CW EPR; 1 G modulation; 40 G scan; 1 mW	https://doi.org/10.1016/0003-9861(91)90023-C	not reported	isotropic one-line signal; no g tensor	sensitivity and initialization	active_model.encounter.g_radical
deltaHpp_SQ	CW EPR	https://doi.org/10.1016/0003-9861(91)90023-C	not reported	unresolved hyperfine and homogeneous/inhomogeneous broadening; cannot uniquely convert to T2	validation only	reference_evidence.hyperfine.aqueous_linewidth_pH7_5
deltaHpp_SQ_replication	CW EPR	https://doi.org/10.1124/mol.115.098798	not reported	unresolved line; not a hyperfine tensor or T2 measurement	validation only	reference_evidence.hyperfine.aqueous_linewidth_replication
A_SQ	EPR; ENDOR; TRIPLE resonance; partial deuteration	https://doi.org/10.1002/mrc.1260260812	not reported in accessible abstract	resolved assignments were investigated; numeric table/tensors/signs/isotope mapping and conditions were not accessible; do not say not investigated	unavailable for parameterization	reference_evidence.hyperfine.exact_study
lambda_0	microwave/optical spectral fit	https://hitran.org/media/refs/HITRAN-1973.pdf	not available in accessible table	gas rovibronic Hamiltonian and convention; never transfer into solution encounter	context only	reference_evidence.oxygen.gas.lambda_0
gamma_0	microwave/optical spectral fit	https://hitran.org/media/refs/HITRAN-1973.pdf	not available in accessible table	gas rovibronic Hamiltonian; never transfer into solution encounter	context only	reference_evidence.oxygen.gas.gamma_0
D_O2_gas	derived convention conversion	https://hitran.org/media/refs/HITRAN-1973.pdf	not propagated	not a new measurement; sign and normalization depend on Hamiltonian convention	context only	reference_evidence.oxygen.gas.derived_D
D_O2_solid_air	94-550 GHz multifrequency EPR	https://doi.org/10.1006/jmre.2000.2175	0.003 cm ⁻¹	frozen solid only; prohibited as solution surrogate	context only	reference_evidence.oxygen.matrix_solid_air

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Panel C. Evidence and permitted use (continued)

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D_O2_N2_matrix	high-field high-frequency EPR	https://doi.org/10.1016/S0921-4526(00)00615-3	not reported	temperature dependent; signal disappears above 25 K; g=2/no rhombicity fit	context only	reference_evidence.oxygen.matrix_nitrogen
T1_O2_solution	proton magnetic relaxation dispersion	https://doi.org/10.1006/jmre.2000.2219	not reported in abstract	bulk fluid scale; no direct time-domain O2 T1; not encounter-specific and not a static g/D/E tensor	sensitivity context	reference_evidence.oxygen.bulk_solution
K_O2_T4L	gas-pressure NMR plus MD	https://doi.org/10.1038/srep20534	7 M ⁻¹	protein-cavity association; not doxorubicin/O2 and not a spin-Hamiltonian value	context only	reference_evidence.oxygen.protein_cavity.association
tau_rot_O2_T4L	molecular dynamics fit	https://doi.org/10.1038/srep20534	0.006;0.02 ps	rapid orientation averaging; not doxorubicin encounter dynamics	context only	reference_evidence.oxygen.protein_cavity.rotational_correlation
g_D_E_O2_encounter	targeted primary-literature search	NA	NA	not found; gas/matrix/liquid/protein values are not substitutes	sensitivity only	reference_evidence.oxygen.doxorubicin_semiquinone_encounter
T1_T2_Tm_SQ	pulse-EPR and linewidth search	NA	NA	only unresolved CW linewidths found; chemical half-life is not spin relaxation	sensitivity only	reference_evidence.hyperfine.spin_relaxation
k_O2_total_pH6	electron pulse radiolysis transient absorption	https://doi.org/10.1038/bjc.1985.74	0.4e8 M ⁻¹ s ⁻¹	total channel only; homogeneous solution; no encounter geometry or spin resolution	quantitative bulk validation	parameter_records.doxorubicin_semiquinone_plus_oxygen_pH6
k_O2_total_pH11_5	electron pulse radiolysis transient absorption	https://doi.org/10.1038/bjc.1985.74	0.2e8 M ⁻¹ s ⁻¹	total channel only; alkaline; no encounter geometry or spin resolution	quantitative bulk validation	parameter_records.doxorubicin_semiquinone_plus_oxygen_pH11_5

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k_O2_total_1981	pulse radiolysis	https://doi.org/10.1016/0003-9861(81)90263-0	not reported	differs from 1985 values; missing conditions prevent reconciliation; never k_D	validation only	parameter_records.d oxorubicin_semiquinone_plus_oxygen_1981
t_half_SQ_anaerobic	pulse radiolysis	https://doi.org/10.1016/0003-9861(81)90263-0	not reported	chemical loss not spin T1/T2 or encounter duration	context only	reference_evidence. encounter.chemical_lifetime_context
diffusion_distance_SQ	calculated from chemical lifetime/diffusion	https://doi.org/10.1016/0003-9861(81)90263-0	not reported	not pair separation or encounter radius; 8 us air-saturated chemical lifetime is not cage time	context only	reference_evidence. encounter.diffusion_distance_context
encounter_geometry_J_D_kassoc_ksep	targeted structure/EPR/kinetics search	NA	NA	not reported in inspected exact-system papers and not found elsewhere	sensitivity only	reference_evidence. encounter.geometry_exchange_dipolar_association_separation_duration
DQ_coherence	targeted transient-EPR; CIDEP/CIDNP; magnetic-field; product-branching search	NA	NA	retrieved exact-system studies did not investigate D/Q observables; no separate direct paper found	sensitivity only	reference_evidence. encounter.coherence
k_D	spin-resolved kinetics search	NA	NA	bulk $M^{-1} s^{-1}$ total rates cannot be converted or decomposed	illustrative only	reference_evidence. encounter.k_D
k_Q	spin-resolved kinetics search	NA	NA	no state-resolved rate or channel yield	illustrative only	reference_evidence. encounter.k_Q
k_Q_over_k_D	spin-resolved kinetics search	NA	NA	no defensible physical range or ordering; 0/0.01/0.1/1 are inactive illustrative benchmarks only	illustrative only	reference_evidence. encounter.k_Q_over_k_D

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Panel C. Evidence and permitted use (continued)

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pKa_HO2	pulse radiolysis	https://doi.org/10.1021/j100711a009	0.1	buffer absence explains disagreement with earlier values; not a membrane/protein pKa	quantitative aqueous	reference_evidence.downstream.HO2_pKa
k_HH	pulse-radiolysis decay	https://doi.org/10.1021/j100711a009	not reported	species-resolved; do not call an O2- plus O2- rate	quantitative aqueous	reference_evidence.downstream.k_HO2_HO2
k_HA	pulse-radiolysis decay	https://doi.org/10.1021/j100711a009	not reported	requires explicit acid-base speciation	quantitative aqueous	reference_evidence.downstream.k_HO2_O2 minus
k_AA	pulse-radiolysis bound	https://doi.org/10.1021/j100711a009	not reported	2026 reevaluation gives tighter <0.3 M ⁻¹ s ⁻¹ ; common neutral-pH effective rates are not this elementary step	quantitative aqueous	reference_evidence.downstream.k_O2minus_s_O2minus
k_SOD_oxidation	pulse radiolysis optical 650/300 nm	https://doi.org/10.1021/ja00790a007	0.2e9 M ⁻¹ s ⁻¹	exact enzyme/preparation only; surrogate for human or mitochondrial SOD	sensitivity benchmark	reference_evidence.downstream.k_CuZnSO4_D_oxidation
k_SOD_reduction	pulse radiolysis optical 650/300 nm	https://doi.org/10.1021/ja00790a007	0.4e9 M ⁻¹ s ⁻¹	exact enzyme/preparation only; surrogate otherwise	sensitivity benchmark	reference_evidence.downstream.k_CuZnSO4_D_reduction
k_SOD_overall	pulse radiolysis plus optical/EPR	https://doi.org/10.1042/bj1390049	0.18e9 M ⁻¹ s ⁻¹	not a universal intracellular SOD1/SOD2 constant	sensitivity benchmark	reference_evidence.downstream.k_CuZnSO4_D_overall
v_O2minus_heart_SMP	SOD-inhibitable cytochrome-c reduction	https://doi.org/10.1016/S0021-9258(17)35747-2	0.2;2.7	90 uM dose; exact buffer oxygen and timing unavailable in accessible abstract	quantitative matched validation	validation_datasets.beef_heart_submitochondrial_superoxide
v_H2O2_heart_SMP	catalase-released oxygen	https://doi.org/10.1016/S0021-9258(17)35747-2	0.3	200 uM dose; detection limit and exact conditions not reported in accessible abstract	quantitative matched validation	validation_datasets.beef_heart_submitochondrial_H2O2

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v_O2minus_Ehrlich_microsomes	acetylated cytochrome-c reduction at 550 nm	https://doi.org/10.1155/2019/9474823	0.26;1.43	135 uM dose; initial linear rate; oxygenation not separately reported	quantitative matched validation	validation_datasets.Ehrlich_microsome_superoxide
v_O2minus_Ehrlich_mitochondria	acetylated cytochrome-c reduction at 550 nm	https://doi.org/10.1155/2019/9474823	0.15;0.61	135 uM dose; alkaline and rotenone-blocked; oxygenation not separately reported	quantitative matched validation	validation_datasets.Ehrlich_mitochondrial_superoxide
v_O2minus_Ehrlich_nuclei	acetylated cytochrome-c reduction at 550 nm	https://doi.org/10.1155/2019/9474823	0.05;0.33	135 uM dose; isolated fraction; oxygenation not separately reported	quantitative matched validation	validation_datasets.Ehrlich_nuclear_superoxide
v_H2O2_Ehrlich	catalase-released oxygen; linear 10-30 min interval	https://doi.org/10.1155/2019/9474823	0.48;0.6;0.04	135 uM fractions;400 uM cells; controls undetectable with detection limits not reported; PBS air-bubbled 30 min	quantitative matched validation	validation_datasets.Ehrlich_H2O2
H2O2_PC3	catalase-aminotriazole kinetic assay	https://doi.org/10.1016/j.abb.2005.06.015	4;13 pM	30 min exposure; oxygen fraction not reported; extracellular Amplex Red did not increase; catalase overexpression did not improve resistance	quantitative matched validation	validation_datasets.PC3_intracellular_H2O2
MitoSOX_H9c2	1 h fluorescence	https://doi.org/10.1016/j.bbrc.2007.04.106	0.04;0.08;0.08	10/20/50 uM doses; n=3; oxygen not reported; doxorubicin autofluorescence and nonspecific probe products prohibit absolute calibration	sensitivity only	validation_datasets.H9c2_MitoSOX
negative_5_imino	acetylated cytochrome-c reduction	https://doi.org/10.1155/2019/9474823	1.43;0.15	both 135 uM; supports quinone dependence but not D/Q selectivity	qualitative disconfirming	validation_datasets.quinone_disabled_negative_control

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